

STE(Stil Time Editor) User Manual

STE
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Chapter 1

Introduction

1.1 Audience

The STE User Manual is intended for test and design engineers of integrated circuits.

1.2 Concept

STE is designed to give the test and design engineers a friendly graphical tool for viewing/editing the timing portion of STIL files so the user shouldn't take care for the STIL standard and syntax .

STE is supporting tool for VCDSTIL (Convert E/VCD files to STIL files).

The only use of STE is edit STIL file timing that the user can use as STIL master file with VCDSTIL.

At the end the STIL file can be translated into a test program for various ATE using Agilent ATEGen tool, See ATEGen user manual for more information.

Chapter 2

Invocation

The STE can be controlled by command line parameters.

2.1 Using Command Line Interface (CLI) of STE

2.1.1 Synopsys

STE [-help] [-version] [-timestamp] [-licwait] [STIL-input-files]

2.1.2 Command Line Options Description

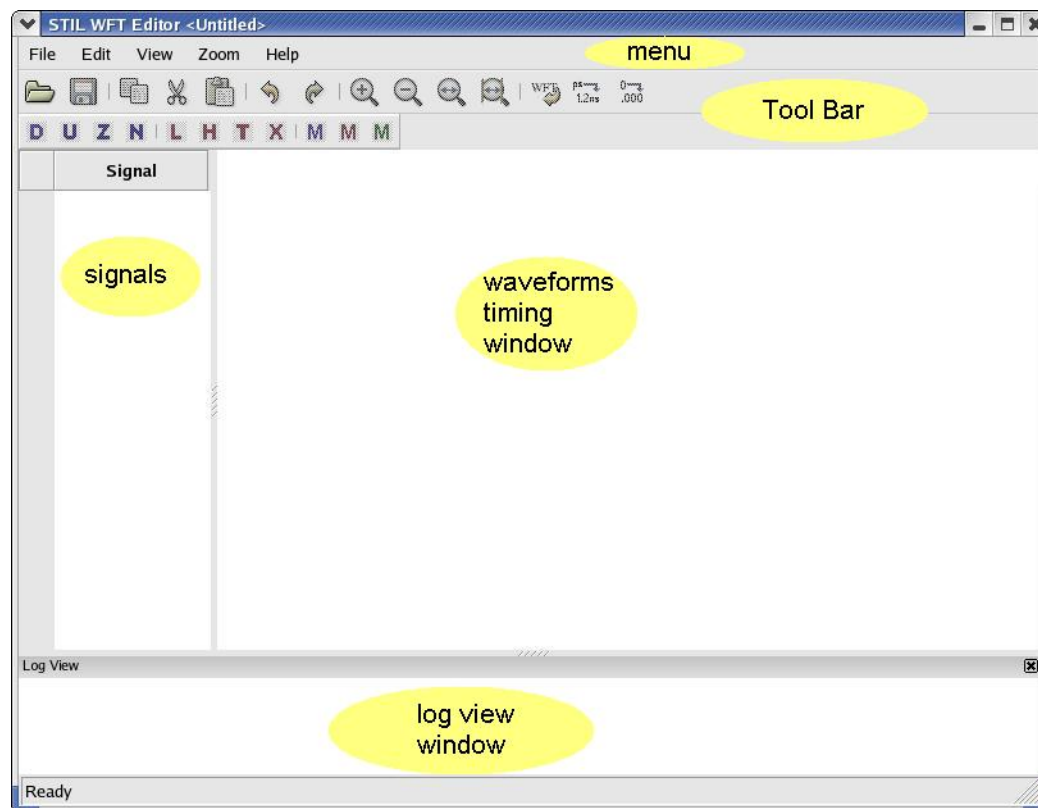
The following are available command line options of STE :

-help	Display help message and exit.
-version	Show version number and exit.
-timestamp	Enable log timestamp.
-licwait	Wait for the license, if not available. The application will wait indefinitely until a license becomes available.
STIL-input-files	Specifies path to STIL file.

Chapter 3

STIL editing with STE

3.1 Screen Explanation



3.1.1 Toolbar Explanation



- Open File



- Save File



- Copy Section



- Cut Section



- Paste Section



- Undo for editing



- Redo for editing



- Zoomin



- Zoomout



- Zoom to fit



- Zoom to fit WFT period, set the screen width the same as the WFT period



- Select WFT, for working with multiple Waveform Tables timing (enabled only when the STIL file has multiple WFT)



- Select units of measurement, possible units: ms, us, ns, ps, fs.



- Select steps of moving signals, possible units: ms, us, ns, ps, fs.

The possible states of the waveforms:



- Input, Force down



- Input, Force Up



- Input, Force off



- Input, Force Unknown



- Output, Compare Low



- Output, Compare High



- Output, Compare Off



- Output, Compare Unknown



- Mask Input



- Mask Output

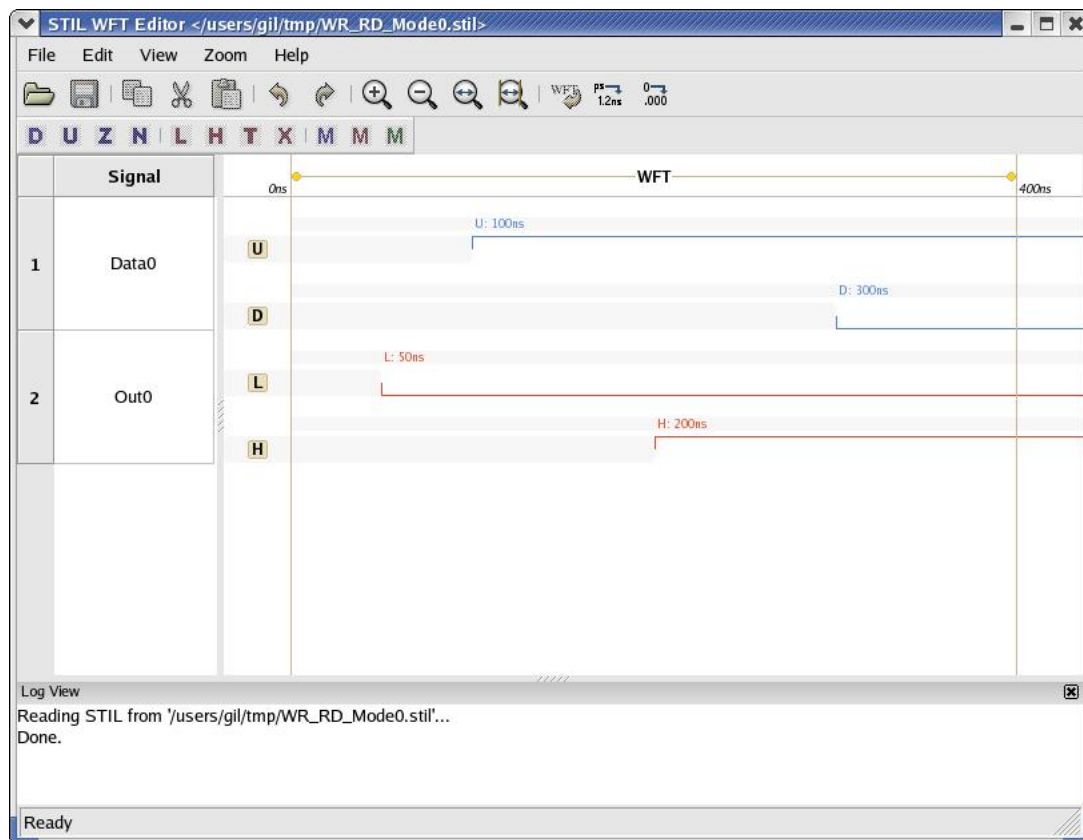


- Mask Any

3.1.2 General view of STIL file

STIL Timing example for pins Data0 and Out0

```
"Data0" {
  U { '100ns' U; }
  D { '300ns' D; } }
"Out0" {
  L { '50ns' L; }
  H { '200ns' H; } }
```



On the left side there is the 'SIGNAL' column which contains all functional signals of the DUT, the right side shows signal waveforms. For example here we have 2 signals, Each Signal has 2 waveforms.

The Blue Color means Input, Red means Output.

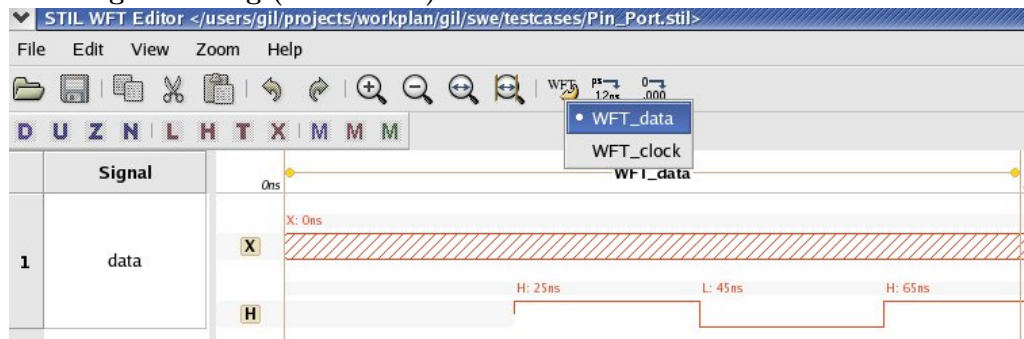
Inout Signals can have both Input and Output waveforms.

3.2 Switching WFT

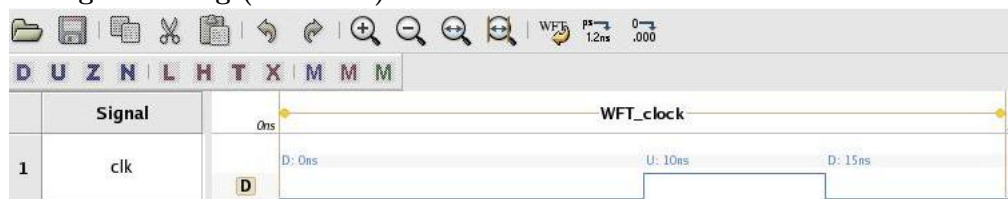
If we have a STIL file with multiple WFT , we can switch from one WFT to another with the button WFT

```
Timing {
WaveformTable "WFT_data" {
    Period '80ns';
    Waveforms {
        "data" {
            X { '0ns' X; }
            H { '25ns' H; '45ns' L; '65ns' H; }}}}
WaveformTable "WFT_clock" {
    Period '20ns';
    Waveforms {
        "clk" {
            D { '0ns' D; '10ns' U; '15ns' D; }}}}
}
```

data signal timing (WFT_data)



clk signal timing (WFT_clk)



3.3 Tips and Tricks

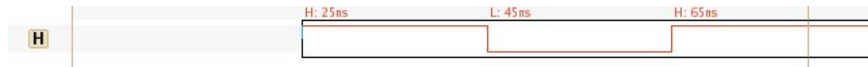
1. The following hot keys are supported in STE :

Copy - CONTROL + C
 Cut - CONTROL + X
 Paste - CONTROL + V

2. Double click on waveform of signal marks the specific event :



3. double click + shift marks all the waveform line

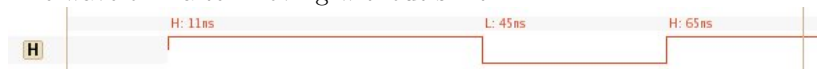


4. In order to move waveform , you can move the whole waveform by press the start of the waveform (the most left side of the waveform) + shift.

The waveform before moving



The waveform after moving without shift



The waveform after moving with shift



3.4 Remarks

1. Remember when you change the STIL waveforms the patterns remain untouched , in order to update the patterns according to the waveforms, you should use the STIL file as STIL master file with VCD2STIL .
2. It is important to understand That all The Mask options : Maskinput, Mask-output and MaskAny are not part of STIL standart, These events don't have any Driving or Comparing meaning! .
the STIL file with these Events wouldn't be able to convert into Tester program, the only use of these events is to mask events in certain time slot when we use the STIL file as master file in VCD2STIL.

Chapter 4

Usage of STE with VCDSTIL

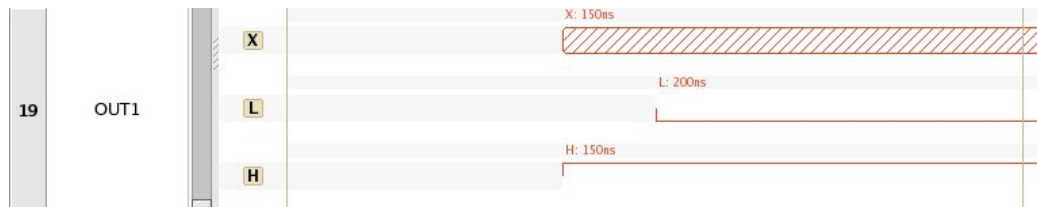
4.1 How to use STIL Master file

In case we want to mask edges of multiple waveforms in the STIL file it is possible to generate STIL file from the VCDSTIL and use it as STIL Master file which would be an input to VCDSTIL .

1. Open STE and load the STIL file.
2. Go to the signal which you want to mask and select the area where you want to mask events.

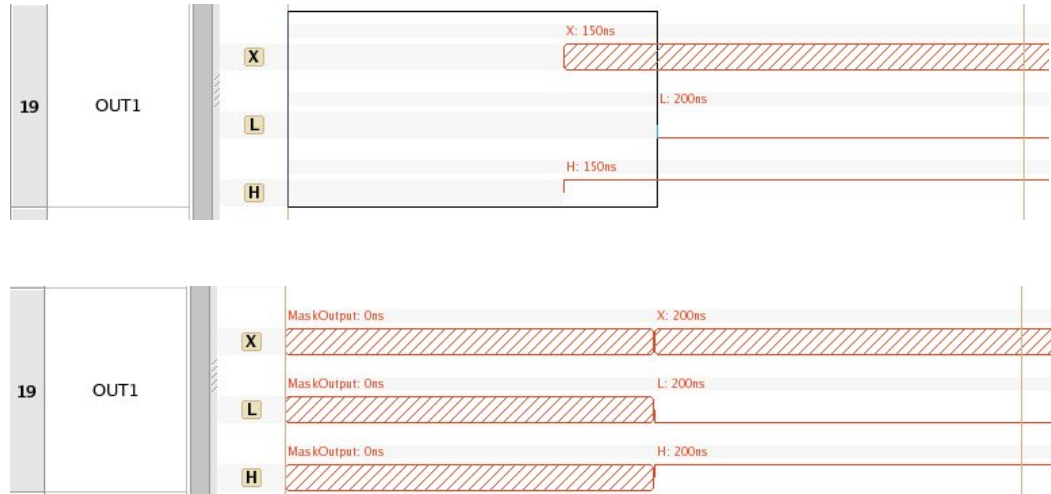
For example here we have a signal OUT1:

```
"OUT1" {  
  X { '150ns' X; }  
  L { '200ns' L; }  
  H { '150ns' H; }  
}
```



We want to mask all the events before 200ns so we would have only one sampling point at 200ns for 'X', 'L' and 'H'.

What we have to do is to mark the area as it shown in the picture and press the 'MaskOutput' button.



The resultant signal in the STIL file:

```
"OUT1" {
X { '0ns' MaskOutput; '200ns' X; }
L { '0ns' MaskOutput; '200ns' L; }
H { '0ns' MaskOutput; '200ns' H; }
}
```

3. After we finish editing the STIL file, save the file under different name and config VCDSTIL to get it as Master file. For more explanation look at the section 'master file' of the VCDSTIL user manual.